ps4: the final project (ps4 will become ps5 if you do really well!)*

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due nov19

1. submit a set (1-10) of maps, typically 3-5, in rare cases 1-2 maps may be enough but only if it is truly outstandingly sophisticated;
2. for presentation, no more than 1 map per minute, so if 5min presentation, no more than 5maps
3. can (and should) reuse stuff from previous ps, just polish, fix-up, make better
4. fa24: less is more! instead of adding more make what you have better; take step back, rethink, interpret

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general directions (always the same):

- i will show your code in class/repost, as per our core values–opensource and transparency, but if for whatever reason you'd like to keep it private, do let me know, its fine, you just may get less feedback
- you must submit all the code that was executed from the very beginning starting with the very raw data as per replication principle; if data too big to fit online, then just start with eg "to fit data online subset to XYZ regions only" or "took 10% random sample" etc–try to keep the files under 25mb so it loads fast
- ps are cumulative–can and should include much of the previous code; can also use code you've written outside of this class (other classes, projects, etc) but you have to clearly mark the code that has not been written for this class–otherwise, scholastic dishonesty! yes, you can plagiarize yourself!
- use your own dataset; again if you do not have a dataset, ask for help finding it
- you are only submitting code, so it must load data from Internet: `https://theaok.github.io/generic/howToPutDataOnline.html` (when you put data into any public space, try not to violate data copyrights... I haven't had anyone having problems with that, but be careful–for instance you may subset dataset to few vars and smaller sample); and it is also easier to learn on small datasets
- keep it simple! at the beginning of your notebook drop unnecessary vars; and even retain only fewer obs; keep it manageable; much easier to learn using simple data; can always complicate later!; much better to do it right using simple data than do it wrong using complex data
- have nice structure in your file: sections, subsections, comments, etc; may also have multiple files
- great to copy code from others; again, one of the rules for this class is 'be lazy': don't reinvent the wheel, whatever you are coding, has already been done, google things often; but of course you cannot submit 100% code by someone's else, and do cite others incl chatGPT, stack overflow, etc
- if you do something extra/fancy that is relevant, it will be extra credit
- use coding rules that we've learned so far
- submit (only) the code into git repo; ps are due by the beginning of the class, ie 6pm; late ps not accepted
- we are on the way to developing the final project with these ps: as we progress, your ps should start resembling a coherent and logical project where you use learned techniques to answer interesting questions– say in few sentences (probably at the beginning) why are you doing what you are doing–that is, answer the "so what question": "ok, you're gonna run all that code, and so what?" what's the goal of all that, why are you doing this? you need a compelling justification for what you are doing: say what are those questions you want to answer; be brief, say couple sentences, and definitely not more than say 100 lines, typically 10-50 lines is enough; related: say why you use data you are using, is it best, does it serve the purpose?; and can ask us questions in comments
- be prepared do present your code in class (if time), just briefly, key points, couple minutes

- if you work in a group of 2 or 3 people make it 2x or 3x better, eg If ps asks for joining 2 datasets, and there are 3 of you, then just join 6, etc, just do 3x more or better
- always have a brief description/interpretation of a map, say few sentences or a paragraph or max few; also may list problems you've encountered and ask questions
- always have exact links to all of the source data (so that i could create the map myself); note: exact links, eg do not say census.gov, but give full url to the data–i must be able o find it; sometimes there is no generic URL–then give steps: what I need to click to get the data!

*If you do really well on this one you are done :) It becomes ps5 and dont need to submit anything else; if there are things to fix up, you fix it up, and submit as ps5